



Algebra 2 covers quadratic equations, conics, polynomials, functions, logarithms, clever factorizations and substitutions, systems of equations, sequences and series, symmetric sums, advanced factoring methods, classical inequalities, functional equations, and more.

This course is specifically designed for high-performing students and draws material from many programs for top middle and high school students in the country. Our philosophy is that students develop more by learning to solve problems they haven't seen before, as opposed to offering repeated drills that students can memorize their way through. In this way, our classes are structured much more like courses at top-tier colleges.

Book: *Intermediate Algebra*, by Richard Rusczyk and Mathew Crawford

Time Commitment: This 36-week course includes 105 minutes in-class and 1-3 hours of homework each week, corresponding to a full year course.

Content:

Week	Topic
1	Solving Equations 1
2	Solving Equations 2
3	Functions
4	Graphing Functions
5	Complex Numbers
6	The Complex Plane
7	Quadratic Equations
8	Graphing Parabolas
9	Inequalities with Quadratics
10	Circles and Ellipses
11	Hyperbolas and Conic Sections
12	Midterm 1
13	Polynomial Division and Factor Theorem
14	Remainder Theorem
15	Integer and Rational Roots
16	Irrational and Nonreal Roots
17	Vieta's Formulas
18	Multivariable Polynomials

Week	Topic
19	Radicals & Radical Functions
20	Arithmetic Sequences & Series
21	Geometric Sequences & Series
22	Infinite Geometric Series
23	More Sequences & Series
24	Midterm 2
25	Basic Counting
26	Binomial Theorem
27	Exponents & Logarithms 1
28	Exponents & Logarithms 2
29	Rational Functions 1
30	Rational Functions 2
31	Special Functions
32	More Special Functions
33	Inequalities
34	AM-GM
35	Induction
36	Final Exam

Common Core State Standards:

Domain	Subdomain	Standards
Number and Quantity	The Complex Number System	7, 8, 9
Algebra	Seeing Structure in Expressions	1b, 3ab, 4
	Arithmetic with Polynomials & Rational Expressions	1, 2, 3, 4, 6, 7
	Creating Equations	2
	Reasoning with Equations and Inequalities	2, 4a, 5, 10, 11
Functions	Interpreting Functions	1, 3, 4, 5, 7bcde, 8b, 9
	Building Functions	1bc, 2, 3, 4abc, 5
	Linear, Quadratic, & Exponential Models	1abc, 2, 3, 4, 5
Geometry	Expressing Geometric Properties with Equations	2, 3

Sample Problems:

- If $a + ar + ar^2 + \dots = 15$ and $a + ar^2 + ar^4 + \dots = 10$, then what is the sum of the geometric series $a + ar^3 + ar^6 + \dots$?

- Let $x = 2^{\log_b 3}$ and $y = 3^{\log_b 2}$. Find $x - y$.

- Show that if a polynomial $f(x)$ leaves a remainder of the form $px + q$ when it is divided by

$$(x - a)(x - b)(x - c),$$

where a, b , and c are all distinct, then $(b - c)f(a) + (c - a)f(b) + (a - b)f(c) = 0$.

- Show that if the roots of $x^3 + ax + b = 0$ are rational numbers m, n , and p , then the roots of $mx^2 + nx + p$ are also rational.

- Show that if a, b , and c are positive, then $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{1}{a\sqrt{bc}} + \frac{1}{b\sqrt{ac}} + \frac{1}{c\sqrt{ab}}$.

- Find the value of

$$\frac{a}{a+1} + \frac{b}{b+1} + \frac{c}{c+1}$$

given that $a + b + c = -1$, $ab + bc + ca = -1$, and none of a, b , and c is equal to -1 .